

Modelling Interaction Dynamics for Behaviour Prediction in Motile, Physical Multi-Agent Systems (Figures Appendix)

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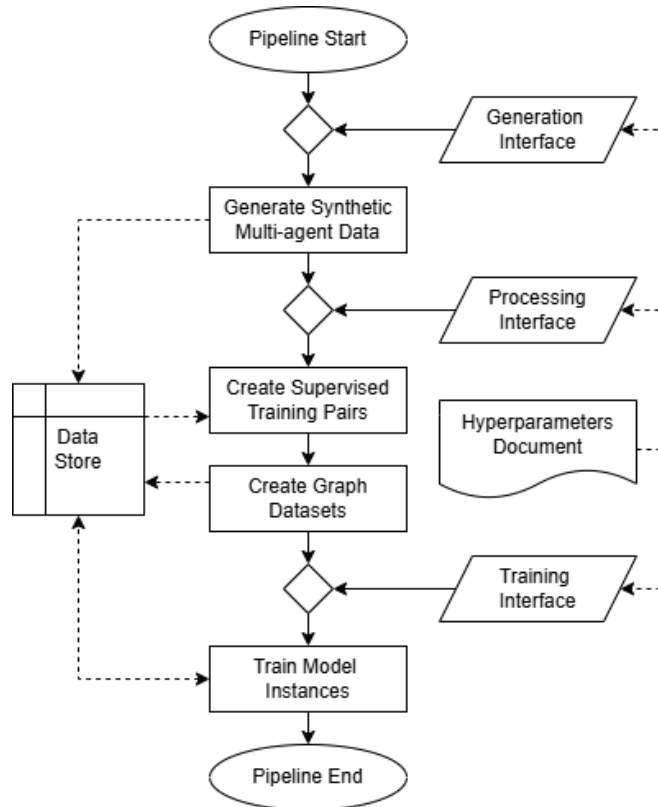


Fig. 1: Pipeline Flowchart

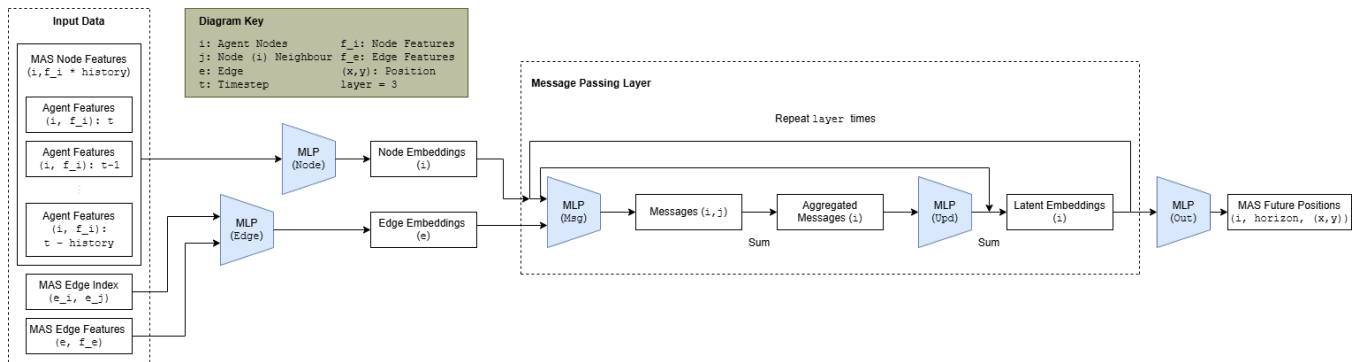


Fig. 2: Approach Architecture Diagram

	Webots	Gazebo	Petting Zoo	VMAS	ARGoS	Isaac Sim
Multi-Agent	Supported	Heavy Setup Overhead	Ideal	Ideal	Ideal	Supported
Scenarios	Limited Scale	Available	Extensive	In-Built	Available	Extensive Set Up Time
Performance	Visuals Over Speed	Fidelity / Performance Tradeoff	Scenario Specific	Vectorised	Focused on Larger Scales	Heavy Hardware Required
Customisation	Modular & Extensible	ROS Integration & Plugins	Modular & Extensible	Modular & Extensible	Requires System - Level Work	Modular & Extensible
Logging	Easy	Difficult	Easy	Easy	Doable	Doable
Fidelity	High Fidelity	High Fidelity	Abstract Physics	Abstract Physics	High Fidelity	Extremely High Fidelity
Friction	Intuitive	Heavy Setup Overhead	Minimal	Minimal	Steep Learning Curve	Steep Learning Curve

Fig. 3: Simulation Platform Against Criteria

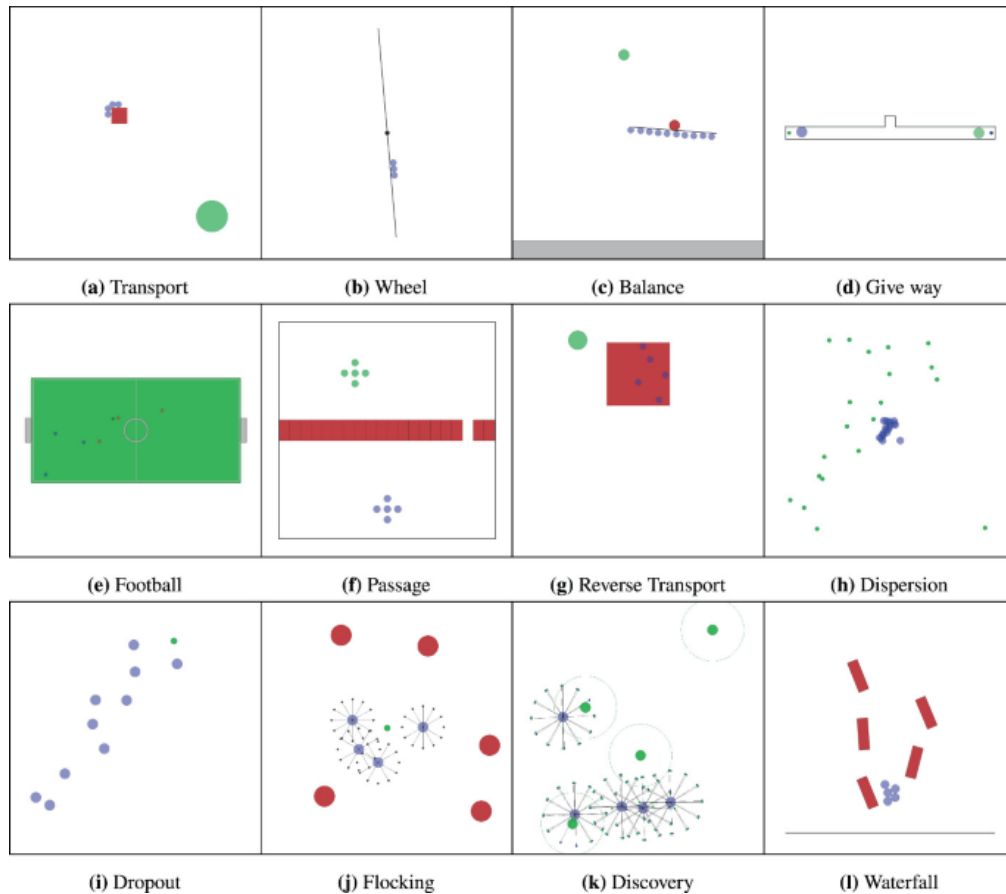


Fig. 4: VMAS Scenarios

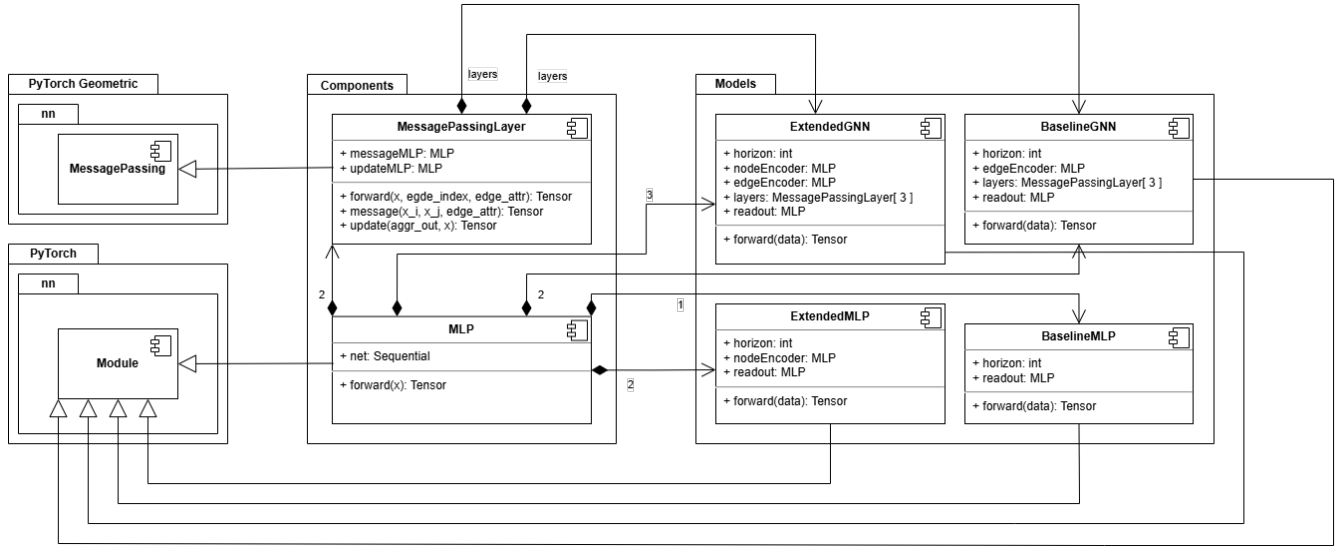


Fig. 5: UML Diagram for Ablation Models Implementation

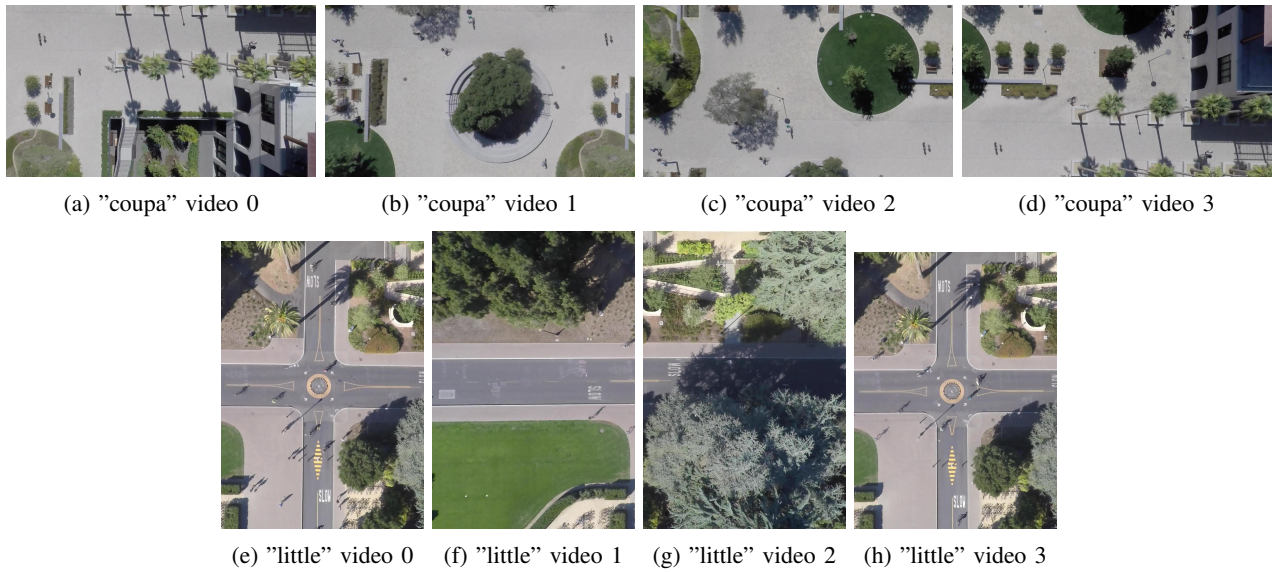
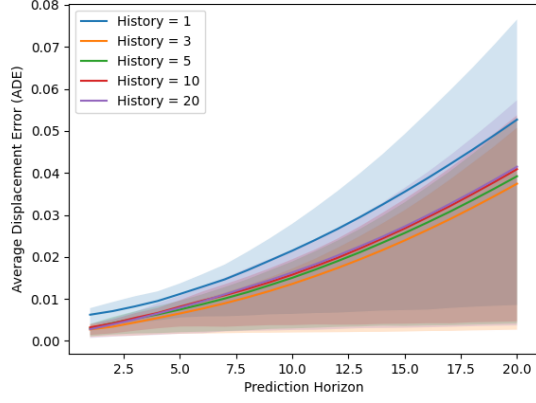


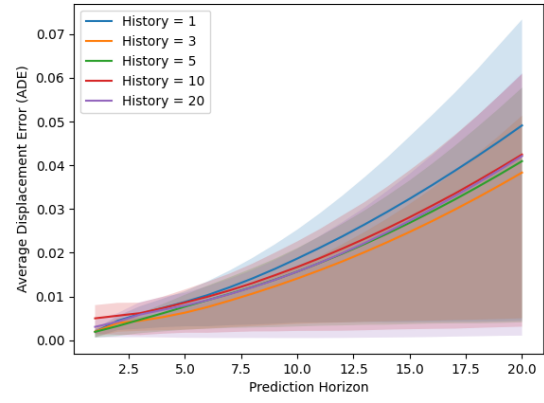
Fig. 6: Human Trajectory Scenarios from the Stanford Drone Dataset (SDD)

Model w/ Message Passing + Temporal Encoder - 'flocking' Scenario (VMAS Simul)



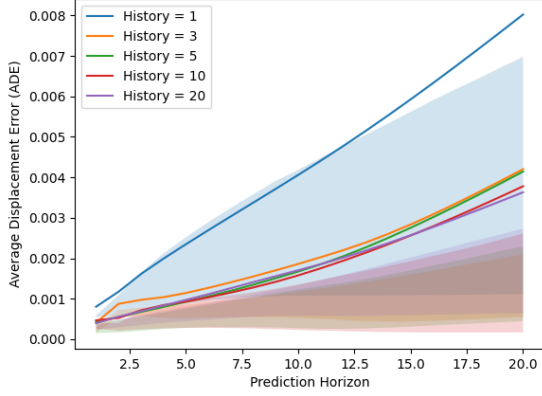
(a) Ext. GNN, Flocking

Model w/ Message Passing Only - 'flocking' Scenario (VMAS Simulation)



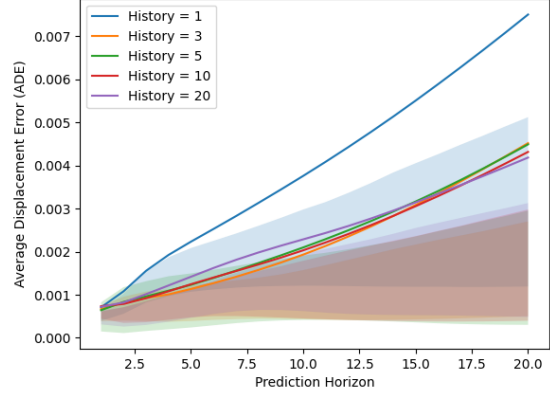
(b) Base. GNN, Flocking

Model w/ Message Passing + Temporal Encoder - 'navigation' Scenario (VMAS Simul)



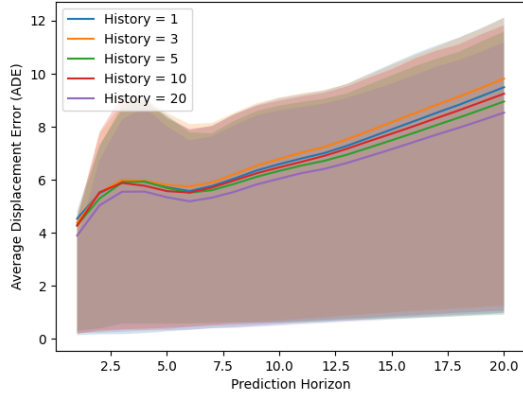
(c) Ext. GNN, Navigation

Model w/ Message Passing Only - 'navigation' Scenario (VMAS Simulation)



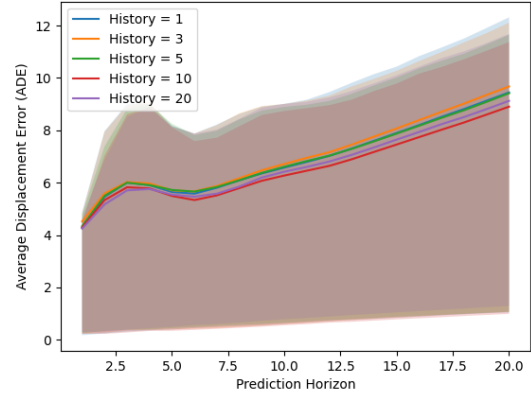
(d) Base. GNN, Navigation

Model w/ Message Passing + Temporal Encoder - 'coupa' Scenario (SDD Dataset)



(e) Ext. GNN, Coupa (SDD)

Model w/ Message Passing Only - 'coupa' Scenario (SDD Dataset)



(f) Base. GNN, Coupa (SDD)

Fig. 7: Model Permutation Performance w.r.t. History & Horizon

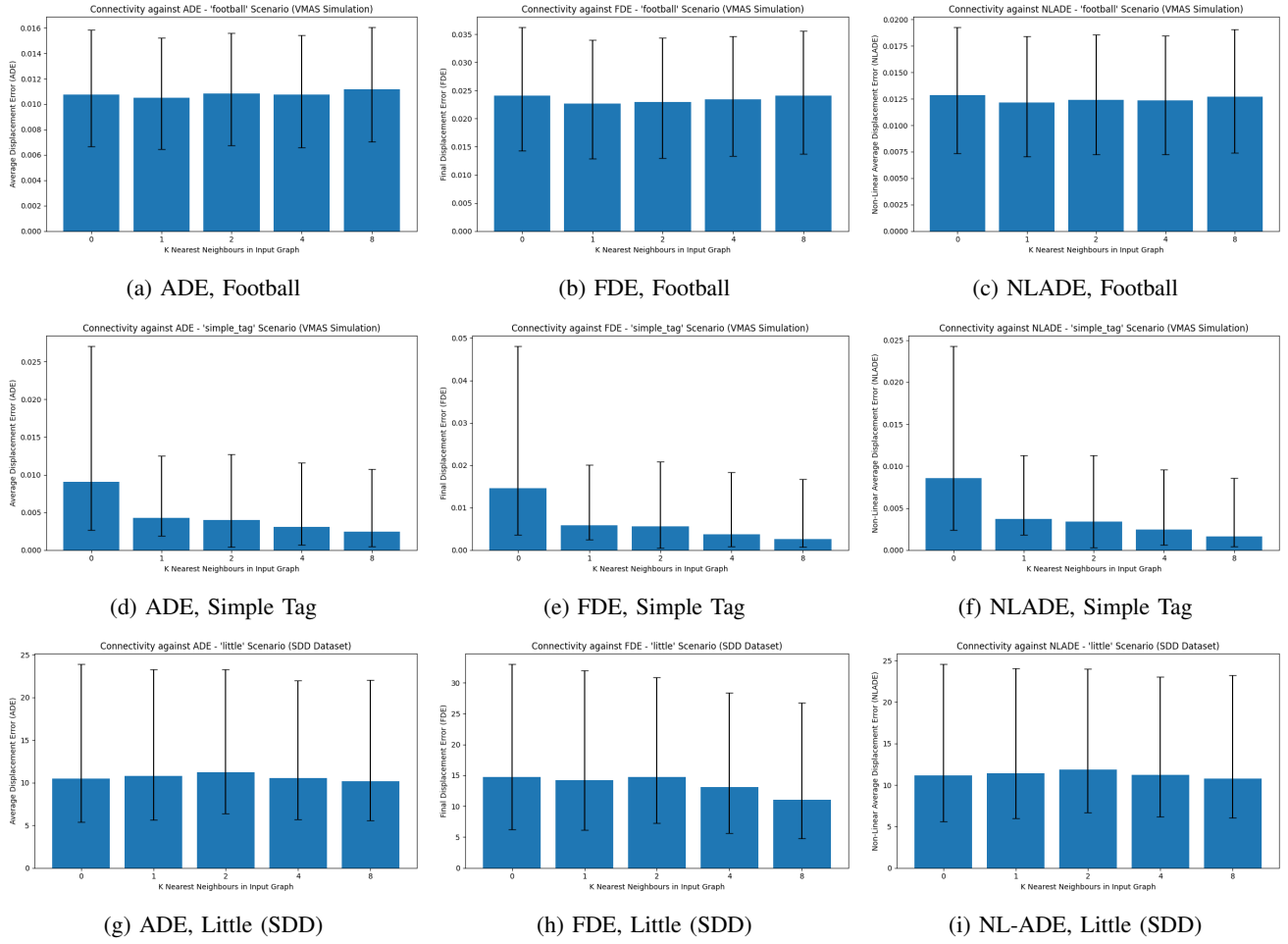


Fig. 8: Model Permutation Performance w.r.t. Connectivity